



2025 Industrial Machinery Benchmark Report

“Service maturity isn’t luck—it’s a strategy. Here’s the proof.”

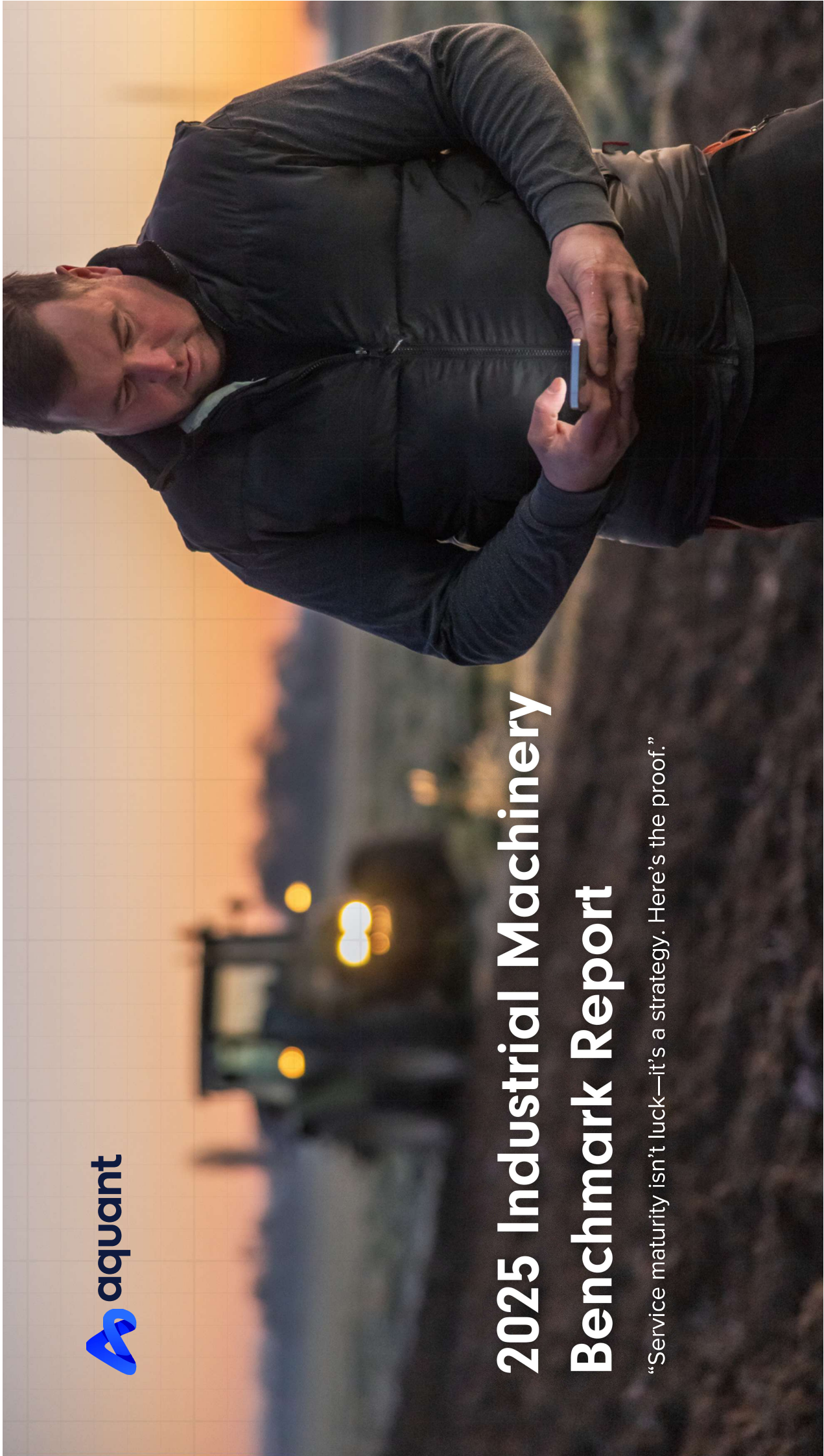


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INTRODUCTION

Industrial machinery service is at a crossroads

New tariffs. Rising costs. Global uncertainty.

In 2025, industrial machinery OEMs are facing a new wave of disruption—this time driven by trade policy. Escalating U.S. tariffs on imported equipment and components have reignited supply chain volatility, sending ripple effects across the sector. Field service organizations that rely on seamless uptime and cost-controlled parts are now grappling with higher sourcing costs, delivery delays, and increasingly thin margins.

For companies with fixed-price service contracts and warranty bundles, the impact is exceptionally sharp. As component costs soar, many are being forced to absorb the difference or reimagine how service is delivered entirely. Parts usage is under scrutiny, particularly for organizations that rely on “shotgunning” (swapping multiple parts to address a single issue). This approach, while sometimes effective, is now more costly than ever, especially when those parts are under tariff pressure.

Worse still, the short-term gain of selling additional parts to customers may come at the expense of long-term satisfaction and loyalty, especially for those locked into contracts that promise value, not surprise expenses. Forward-thinking OEMs are responding by shifting their service models. It's no longer just about dispatching techs or replacing parts. It's about working smarter: extending equipment life through predictive maintenance, remote diagnostics, and guided troubleshooting. **It calls for a new strategy—one grounded in domain-specific intelligence and scalable knowledge.**

The **2025 Industrial Machinery Benchmark Report** highlights how top service organizations are meeting the challenge—and turning it into a competitive advantage. By adopting agentic AI platforms that embed service expertise into every decision, they're closing the Service Expertise Gap, operating with greater precision, and generating real business value.

We analyzed more than **4.8 million service events, spanning 1.6 million assets, across 40 leading industrial machinery companies** to understand what separates the leaders from the laggards. Here's what we found.

How we compiled the data

Aquant gathered and analyzed anonymized data across leading industrial machinery companies.

Note: Data from OEMs with and without dealer networks are included within this set.

This report highlights the performance patterns, blind spots, and opportunities that distinguish the most efficient organizations from those stuck in reactive cycles.



40+
organizations



More than **4.8 million**
service events



Nearly **1.6 million**
assets



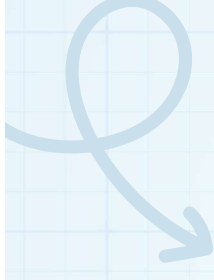
Over **54,000**
technicians

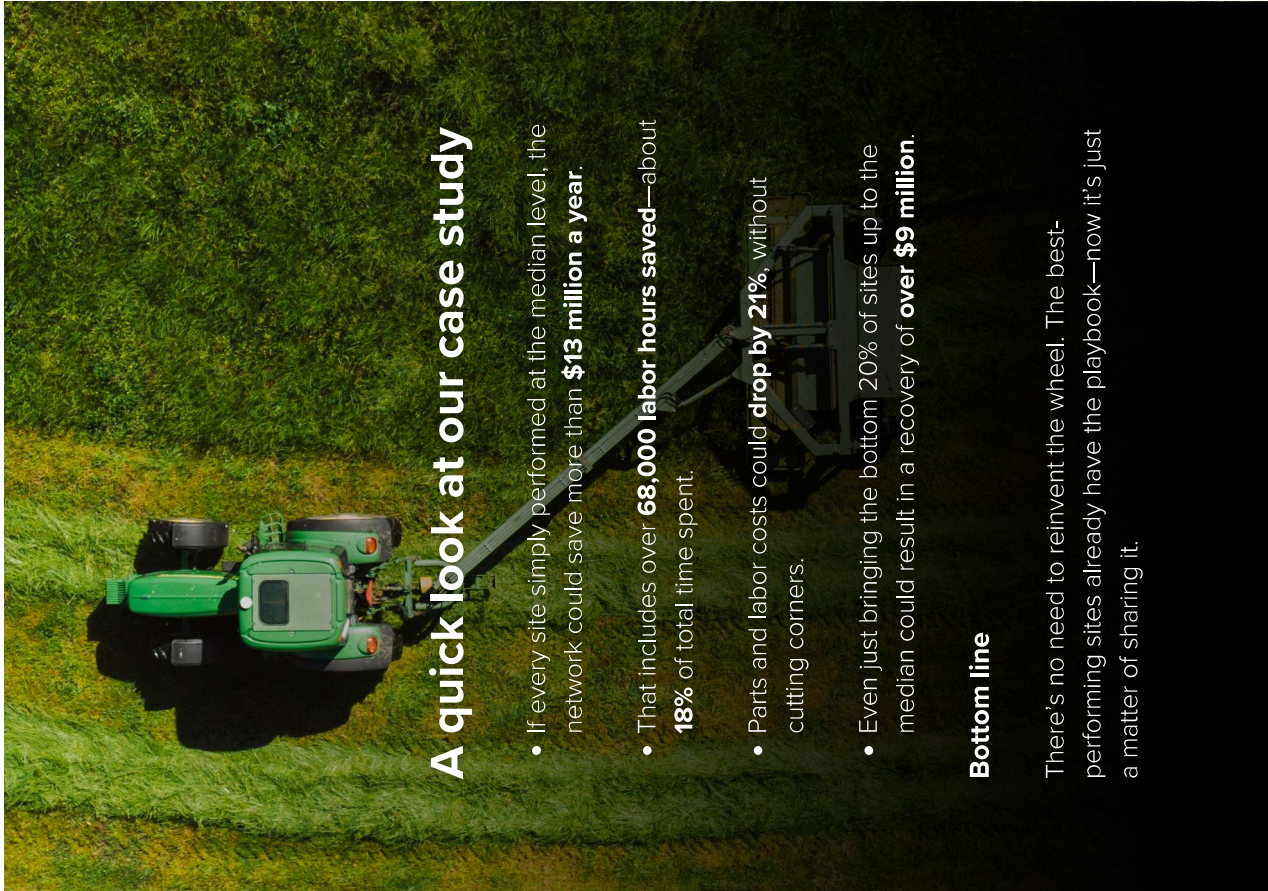


Almost **\$3.5 billion**
in service costs



An average of **3 years of**
service data per company





A quick look at our case study

- If every site simply performed at the median level, the network could save more than **\$13 million a year**.
- That includes over **68,000 labor hours saved**—about **18%** of total time spent.
- Parts and labor costs could **drop by 21%**, without cutting corners.
- Even just bringing the bottom 20% of sites up to the median could result in a recovery of **over \$9 million**.

Bottom line

There's no need to reinvent the wheel. The best-performing sites already have the playbook—now it's just a matter of sharing it.

Key Findings

- **Knowledge = ROI:** Top organizations achieve an 87% First Time Fix Rate, while bottom performers lag at 53%. That's more than just a statistic; it reflects how well (or poorly) organizations capture, scale, and apply expert knowledge across their workforce.
- **Speed is a brand differentiator:** Top performers resolve issues in just two days, compared to nine days for the bottom tier. Long delays damage trust, increase downtime costs, and hurt brand reputation.
- **Frequent visits = reactive service:** A failed first visit adds 13 extra days and typically requires two additional visits to resolve the issue. Top organizations average 123 days between visits, while bottom organizations revisit assets every 48 days, often a sign of misdiagnosis, repeat failures, and unresolved root issues.
- **Cost exposes operational maturity (and the Service Expertise Gap is also a financial gap):** Bottom performers spend over eight times more per service event than top organizations.

These inflated costs reflect inefficient processes, repeat work, and poor triaging. Bottom organizations could lose up to \$3.5 million annually due to low-performing employees. Elevating all techs to top-performer levels could reduce service costs by 31%.

- **Preventative maintenance alone isn't enough:** At bottom-performing companies, 1 in 4 assets break down within 3 months of a preventive maintenance visit—compared to just 3% at top performers. AI-guided insights and contextual support are crucial to transforming PM into a reliable strategy.
- **Agentic AI and domain-specific platforms are the game-changers:** Top performers aren't just digitizing—they're adopting intelligent systems that route each question to the right task-specific agent, embedding expert knowledge into every interaction.
- **Platform strategy matters:** Building fragmented in-house tools or relying on horizontal AI platforms leads to uneven results. The most successful organizations are blending AI approaches within domain-specific platforms designed to address the complexities of industrial services.

Critical KPIs

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First Time Fix Rate (FTFR)

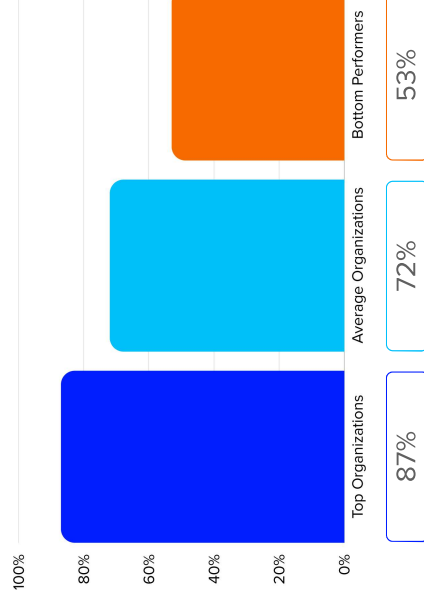
The First Time Fix Rate gap exposes a bigger problem: That 34-point difference isn't just a performance metric—it's a mirror of how well organizations can capture and scale expertise. As seasoned technicians retire, bottom performers struggle to fill the gap, leading to misdiagnoses, repeat visits, and higher costs.

Top performers do three things differently:

- **They operationalize expertise** by embedding it into their workflows via agentic AI platforms—so techs are guided by the right steps, not just tribal knowledge.
- **They democratize decision-making**, providing every technician with access to contextual, real-time support, including those who are newer or less experienced.
- **They resolve more issues earlier**, using AI to triage and handle lower-level problems remotely or at the point of first contact—freeing up field resources for complex work.

Why it matters

When you fix the issue the first time, you reduce repeat visits, minimize downtime, and earn customer trust. But if your First Time Fix Rate is just 53%, nearly half of your service calls require a return trip. That doesn't just drain resources—it undermines your reputation and bottom line.



Resolution Time

Speed is the new service standard: Top performers resolve issues in just two days, **while bottom performers take four times longer**. In a world where uptime is mission-critical, this delay is not just inconvenient—it can cost customers millions in lost productivity, SLA penalties, and customer churn.

What top performers are doing differently:

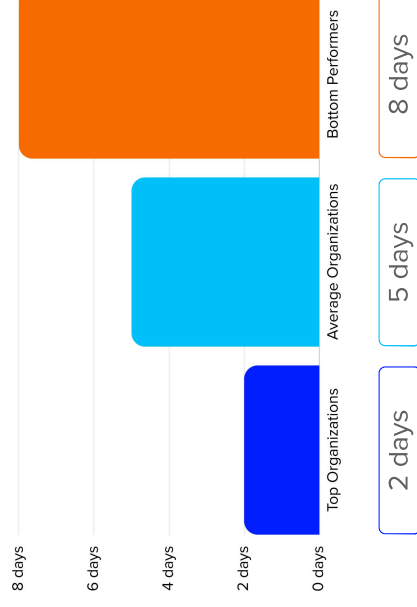
- They **use agentic AI** to route each issue to the right tech with the right context—eliminating back-and-forths and reducing handoffs.
- They **rely on predictive insights** to catch issues before they escalate.
- They **connect systems and teams through domain-specific platforms** that surface the right data and provide next steps in real time.

Bottom line

Fast fixes earn loyalty. Slow ones signal chaos—and give your competitors a foot in the door.

Pro Tip

On average, a failed first visit results in **three total visits** to resolve the service issue and adds approximately **13 days** to your Resolution Time.



Top performers resolve issues
4X faster



3

Time Between Service Visits

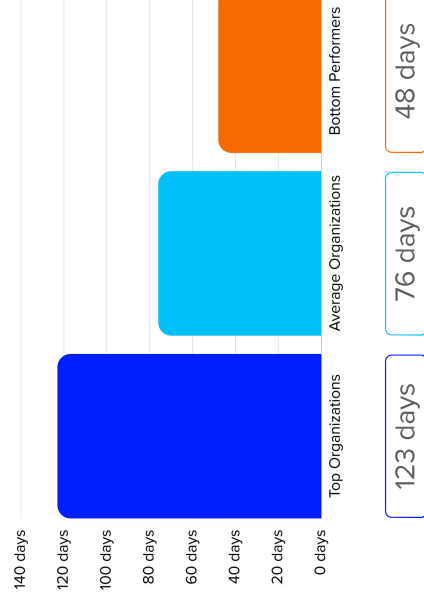
Fewer visits, stronger performance: A longer Time Between Service Visits—such as the 123-day gap in top organizations—signals that machines are running reliably, issues are being resolved thoroughly, and preventive maintenance is effective. On the other hand, bottom performers revisit assets every 48 days, chasing unresolved problems or finally fixing what wasn't fully diagnosed.

Why the gap exists:

- **Bottom performers often lack root-cause resolution**—techs patch symptoms instead of fixing underlying issues.
- Without AI guidance or access to service history, **teams rely on trial-and-error**, leading to more visits, higher costs, and lower customer confidence.
- **Top performers use agentic AI to guide accurate repairs**, detect early signs of failure, and extend asset uptime through tailored recommendations.

Every extra visit adds cost and complexity. It's not just about technician hours—it's increased travel, higher parts usage, more

downtime, and lower productivity. Fewer, smarter visits are the mark of a mature, AI-augmented service model.



1 in 4

assets break down within three months of a PM visit

Did you know? Preventative maintenance isn't working for everyone!

Bottom-performing organizations see one in four assets break down within three months of a PM visit—a huge red flag. That means their PM programs are either not addressing root issues, missing critical failure indicators, or being performed inconsistently/without the right data.

Resolution Cost

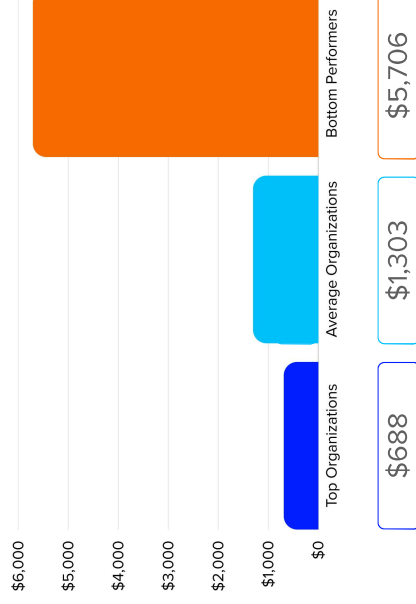
The real cost of inefficiency: Bottom performers spend **eight times more** per service event. The drivers? Low First Time Fix Rates, repeat truck rolls, incorrect part replacements, and long resolution times. This suggests that these organizations are operating in a constant firefighting mode, **where every fix is more expensive than necessary**.

Top performers operate with precision:

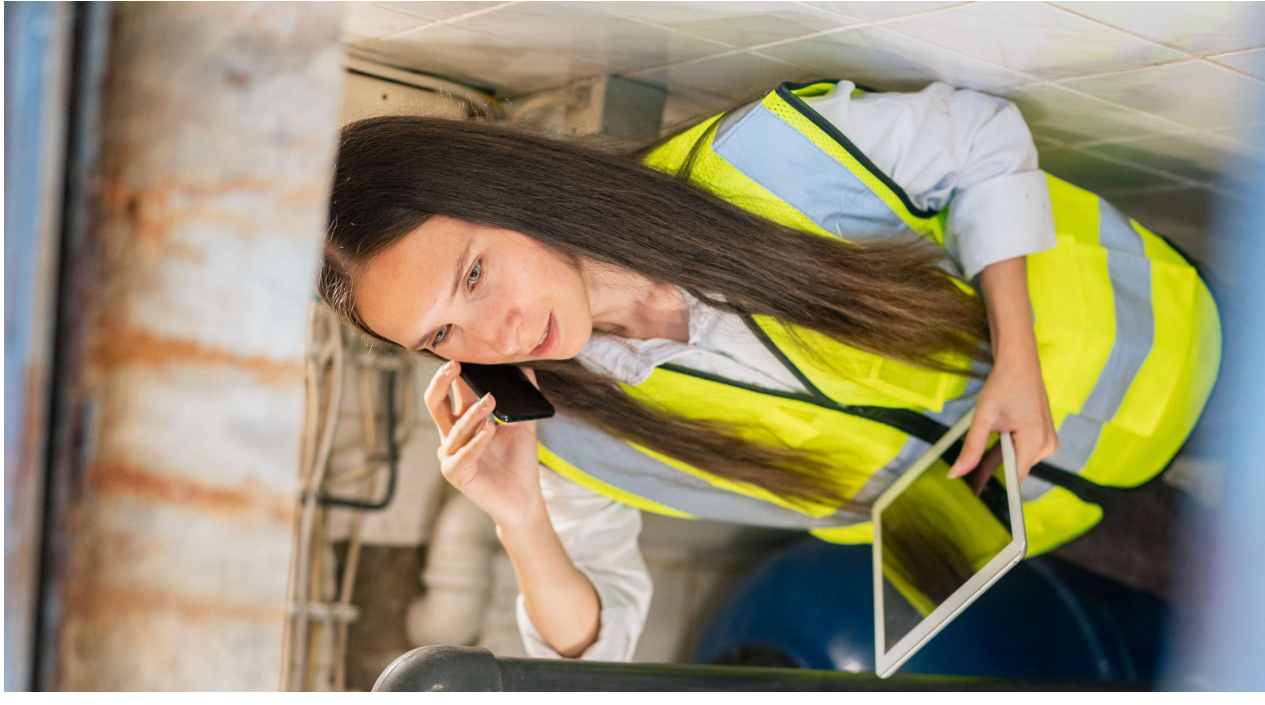
- They use agentic AI to route each problem to a specialized expert agent—delivering fast, context-aware guidance.
- They empower every technician with on-the-job decision support that reduces guesswork and prevents trial-and-error fixes.
- They fix more problems remotely, avoid unnecessary part replacements, and reduce the need for costly dispatches.

Service tools protect margins, increase capacity, and drive revenue: With the right AI platform, service teams don't just save money. They increase profitability by resolving issues more quickly, minimizing waste, and identifying upsell

opportunities, such as warranty extensions or maintenance contracts. It also enables them to handle more service events and sell more parts with the same workforce.



Smart spending beats high spending. These cost deltas aren't about who's cheap or premium. They reflect who's running a modern, intelligent service operation—and who's stuck in a reactive, inefficient cycle.





The Service Shift: How top OEMs are transforming

What separates top industrial machinery organizations isn't just tenure or headcount—it's strategy. Service leaders are winning by:

- **Resolving issues earlier with agentic AI:** Top teams use intelligent platforms that route each question to the right expert agent—enabling remote resolution, smarter troubleshooting, and self-service where appropriate. This reduces escalations and frees skilled technicians to focus on high-value work.
- **Closing the Service Expertise Gap with embedded knowledge:** By capturing institutional know-how and surfacing it in real-time, domain-specific AI helps onboard new techs faster, standardize service quality, and reduce reliance on tribal knowledge.
- **Investing in clean, connected data:** Leaders are unifying service data across ERPs, CRMs, and dealer networks—creating a single source of truth that feeds smarter AI recommendations and gives teams full visibility into asset and performance history.

- **Moving beyond traditional KPIs:** Instead of tracking activity alone, they're measuring what actually drives business value—cost of inefficiency, repeat visit rates, service profitability, and AI engagement.

Pro Tip

If everyone had the knowledge and skills to perform like the top 20% of the workforce, service costs would be reduced by as much as **31%**. At bottom-performing companies, low-performing employees can cost you up to **\$3.5 million per year**.

Low performing employees
can cost you up to
**\$3.5 million
per year**

CASE STUDY

The \$13M opportunity hiding in plain sight

In industrial service, performance isn't just about doing the job—it's about doing it efficiently, consistently, and at scale.

But what happens when service performance varies dramatically across different sites, even within the same network?

That's exactly what we uncovered in our latest benchmark analysis of a significant industrial machinery brand's service data. We examined hundreds of dealerships, ranging from high performers to those lagging behind, and identified a considerable gap with substantial financial implications.

By comparing service metrics—such as labor hours, part usage, and total cost—across the top 20%, the average, and the bottom 20% of companies, **we observed a pattern that's impossible to ignore.**

+\$13 million in combined labor and parts costs savings

Pro Tip:

Even bringing the bottom 20% of dealerships up to median performance could recapture **over \$9 million annually**, with significant savings in both parts and labor.

Pro Tip:

Use AI to standardize what works. Identify which teams excel by equipment model, isolate what they're doing differently, and distribute that know-how through guided workflows, training, and intelligent support systems.

The findings

If every dealership could match **even the median level of performance**, the network as a whole would stand to annually save:

- Over 68,000 labor hours annually—equal to an **18% reduction in labor time**
- More than **\$13 million** in combined labor and parts costs
- A **21% reduction in total service costs**

What's driving the gap?

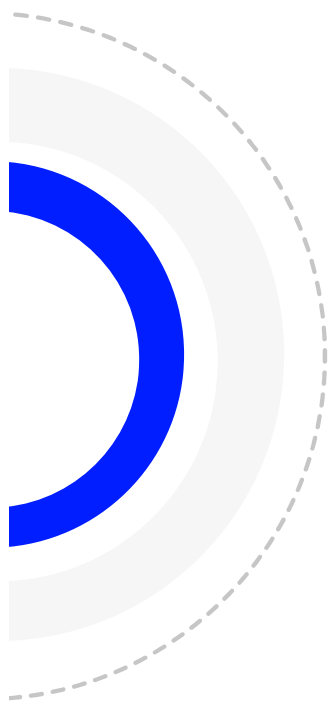
Top performers resolved issues faster, avoided unnecessary part replacements, and required fewer visits to complete jobs. That tells us the challenge isn't necessarily about technician skill or available resources, but rather, that knowledge wasn't being shared or scaled.

Agentic AI closes the loop

To drive network-wide performance, organizations must capture the service intelligence that already exists and make it usable everywhere. Agentic AI platforms make this possible by:

- Identifying what top performers are doing right
- Capturing those steps and decision patterns
- Embedding that expertise into real-time guidance for every tech across the network

The greatest opportunity isn't a new tool or more headcount—it's **scaling what already works**. By unlocking and distributing internal best practices and expert knowledge through a domain-specific, agentic AI platform, OEMs can reduce cost, increase consistency, and elevate performance across every dealership and every technician.





Your next steps

Service excellence doesn't happen by accident—it's engineered. Whether you're just beginning to digitize or advancing an AI-first strategy, this year's data makes one thing clear: closing the Service Expertise Gap is the fastest path to operational efficiency, customer loyalty, and revenue growth.

So, what should your focus be?

- **Scaling what already works:** The knowledge your top performers use every day can—and should—be embedded across your workforce. Agentic AI platforms make this possible by capturing expert insights and routing them to the right people at the right moment.
- **Resolving more intelligently:** Use AI to triage accurately, reduce dispatches, and guide techs in real time. The more issues you resolve early, whether that's through remote support, self-service, intelligent decision-making, or beyond, the more you protect margins and improve KPIs.
- **Rethinking your AI strategy:** Building tools in-house may offer control, but it rarely scales. Horizontal platforms lack context. Top-performing service teams are adopting blended, domain-specific platforms purpose-built for the complexity of servicing industrial machinery.
- **Turning every service event into a learning opportunity:** High-performing organizations treat every job as a data point—refining workflows, improving preventative maintenance schedules, and enhancing AI recommendations over time.

Ready to see how you stack up?

Dive into interactive charts and benchmarks across key metrics like First-Time Fix Rate, Resolution Time, and Cost per Work Order. Filter by performance tier to uncover where you stand—and where you can improve.

[View the dashboard](#)

Agentic AI for professionals servicing complex equipment

Join our team of service experts and learn how you can deploy pre-built agents, build custom ones, or integrate your existing AI, all on one purpose-built platform for complex service operations.

[Talk to us](#)

About Aquant

Aquant is the Agentic AI platform purpose-built for service organizations that maintain complex equipment. Unlike generic AI tools, Aquant combines domain expertise with the power of intelligent agents, enabling teams to build, integrate, and deploy AI Agents for any challenge, across any channel.

The platform captures both documented knowledge and the institutional expertise of service professionals, ensuring that field technicians, contact center agents, service leaders, and customers get expert-level guidance tailored to every job, user, machine, or task.

By analyzing service data—from machine history and service manuals to technician notes and real-world repair patterns—Aquant improves troubleshooting, accelerates workforce training, reduces costs, upsells service contracts, and turns service operations into a powerful revenue engine.

With Aquant AI, service teams are **ready for anything**.

